

Data and Computer Communication

Question Bank – Modules 3, 4 & 5

Module 3 – Digital Transmission and Analog Transmission

3 Mark Questions

- 1 What is digital data to digital signal conversion?
- 2 Explain Non-Return-to-Zero (NRZ) encoding.
- 3 What is Return-to-Zero (RZ) encoding?
- 4 What is Biphase encoding?
- 5 What is Multilevel binary encoding?
- 6 State the Sampling Theorem.
- 7 What is Pulse Code Modulation (PCM)?
- 8 Write the steps involved in PCM.
- 9 What is Delta Modulation (DM)?
- 10 Define Amplitude Shift Keying (ASK).
- 11 What is Frequency Shift Keying (FSK)?
- 12 Define Phase Shift Keying (PSK).
- 13 What is Amplitude Modulation (AM)?
- 14 Define Frequency Modulation (FM).
- 15 Define quantization error in PCM. How can it be reduced?
- 16 Draw the constellation diagram for 4-PSK (QPSK).

Essay Questions

- 1 Explain Biphase encoding schemes (Manchester and Differential Manchester).
- 2 Explain the Nyquist Sampling Theorem and Aliasing.
- 3 Explain Phase Modulation (PM) and how it differs from Frequency Modulation (FM).
- 4 Discuss Multilevel Binary encoding schemes (AMI and Pseudo ternary).
- 5 Represent the data stream 10100111 using NRZ-L, NRZ-I and Manchester encoding.
- 6 Explain Pulse Code Modulation (PCM) with a block diagram.
- 7 Explain Delta Modulation and its quantization errors.
- 8 Explain the generation of ASK and FSK with diagrams.

Module 4 – Multiplexing and Spread Spectrum

3 Mark Questions

- 1 Define Multiplexing.

- 2 What is Frequency Division Multiplexing (FDM)?
- 3 Define Wavelength Division Multiplexing (WDM).
- 4 Define Time Division Multiplexing (TDM).
- 5 What are the characteristics of multiplexing?
- 6 What is Synchronous TDM?
- 7 What is Statistical TDM?
- 8 What is Spread Spectrum?
- 9 What is a chipping sequence in Spread Spectrum?
- 10 Explain the purpose of guard bands in FDM.
- 11 Differentiate between Synchronous TDM and Statistical TDM.

Essay Questions

- 1 Explain Synchronous Time Division Multiplexing with a diagram.
- 2 Explain Frequency Hopping Spread Spectrum (FHSS) and its advantages.
- 3 Explain how WDM is similar to FDM but used in fiber optics.
- 4 Discuss the properties of Pseudo Noise sequences used in CDMA.
- 5 Explain the FDM multiplexing and demultiplexing process.
- 6 What is Spread Spectrum? Explain Direct Sequence Spread Spectrum (DSSS).
- 7 Compare FDM and TDM.
- 8 Explain Code Division Multiple Access (CDMA) with an example.

Module 5 – Error Detection, Correction and Switching

3 Mark Questions

- 1 Define Asynchronous transmission.
- 2 What is Synchronous transmission?
- 3 What are types of errors in data communication?
- 4 Distinguish between Even Parity and Odd Parity.
- 5 What is Checksum?
- 6 Define Cyclic Redundancy Check (CRC).
- 7 What is Forward Error Correction (FEC)?
- 8 What is Hamming Distance? Calculate it for 10101 and 11110.
- 9 What is Message Switching?
- 10 Explain encapsulation in packet switching.

Essay Questions

- 1 Calculate the Checksum for data units 10101010 and 00110011.
- 2 Compare Circuit Switching and Packet Switching.
- 3 Explain Asynchronous and Synchronous transmission modes.

- 4 Using Hamming Code (7,4), encode the data 1010.
- 5 Generate CRC for data word 1010011110 using divisor 1011.
- 6 Explain Circuit Switching and the three phases of communication.
- 7 Explain Datagram and Virtual Circuit approaches in packet switching.
- 8 Explain Hamming code for error correction and calculate parity bits for data 1001.